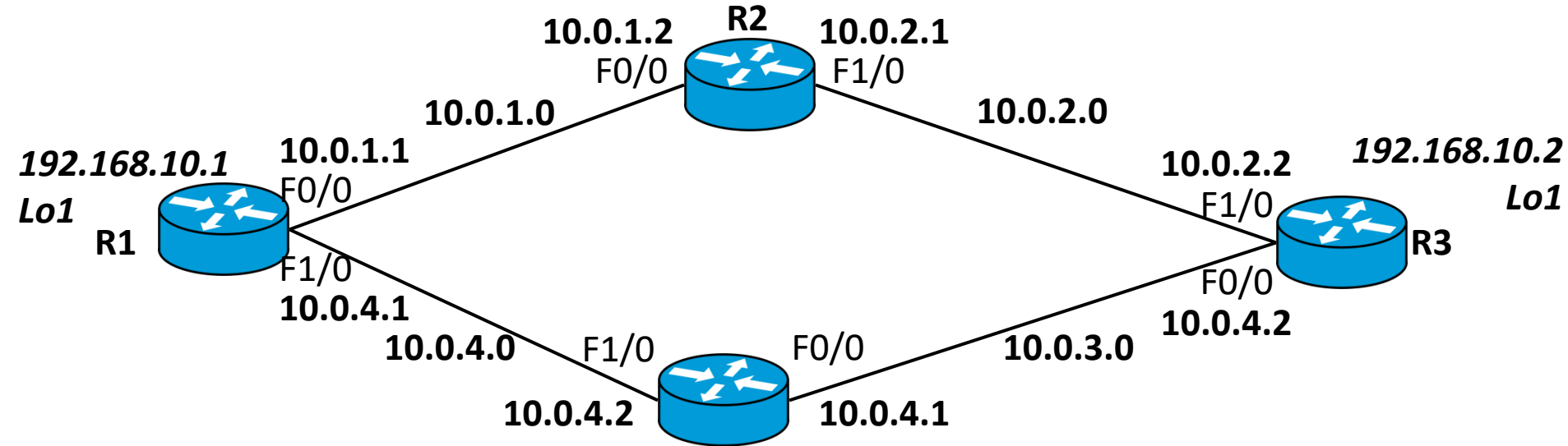


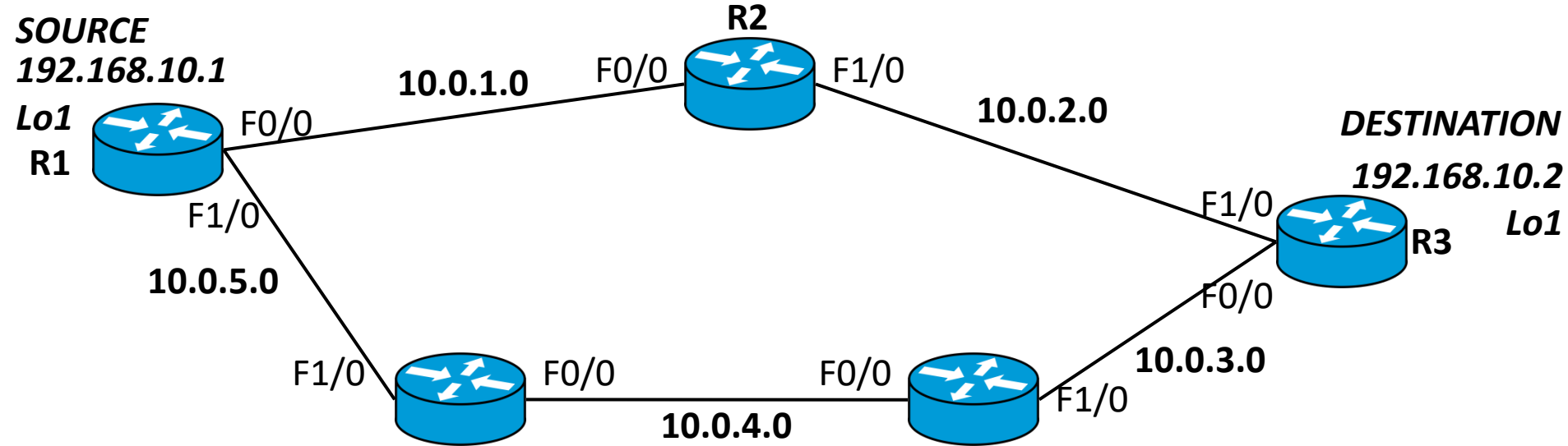
Best Path in EIGRP:



Explanation:

Let us assume Source is 192.168.10.1 and Destination is 192.168.10.2
Here load balancing will occur and best path will be none in this case so both will be best path because route metric is same for both.

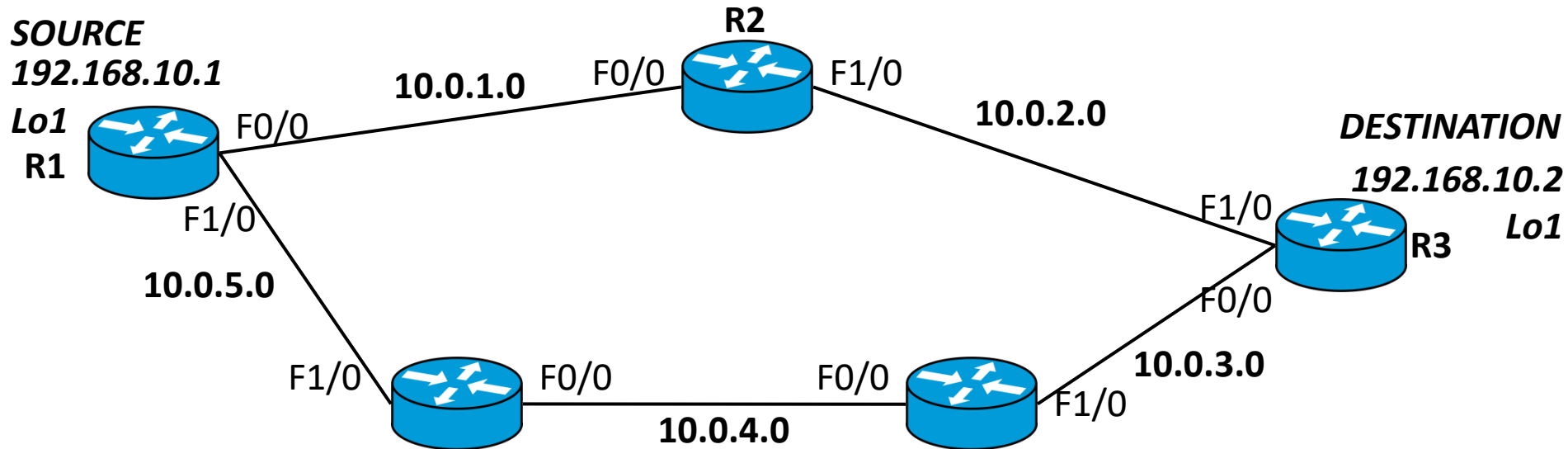
Best Path in EIGRP Example:



Explanation:

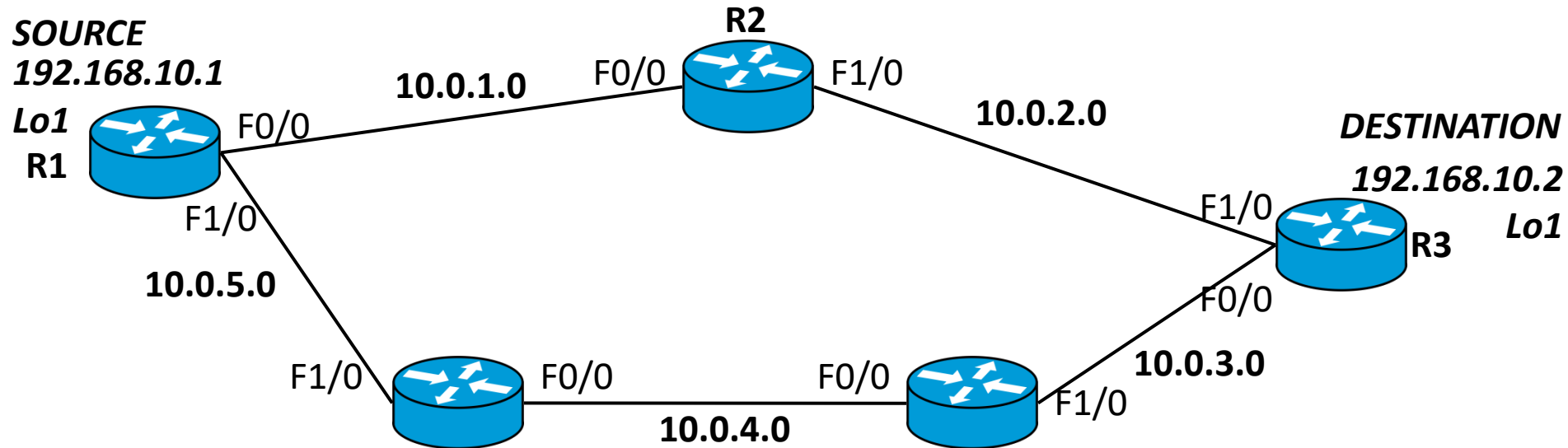
Let us assume Source is 192.168.10.1 and Destination is 192.168.10.2. Here we got two path to forward the data in which we have to find the Route Metric for both Path. And finally the path having low route metric will be automatically chosen as best path to forward data

Delay & B/W of Fastethernet and Loopback (Path1) in EIGRP



All fast ethernet ports has delay and B/w of 100 us and 100000 kb/s.
Loopback interface on R3 has a delay of 5000 us and 8000000 kb/s
Total Delay for Path 1 is 5200/10 us or 520 and Least Bandwidth is 100000 kb/s

Delay & B/W of Fastethernet and Loopback (Path1) in EIGRP



All fast Ethernet ports has delay and B/w of 100 us and 100000 kb/s.

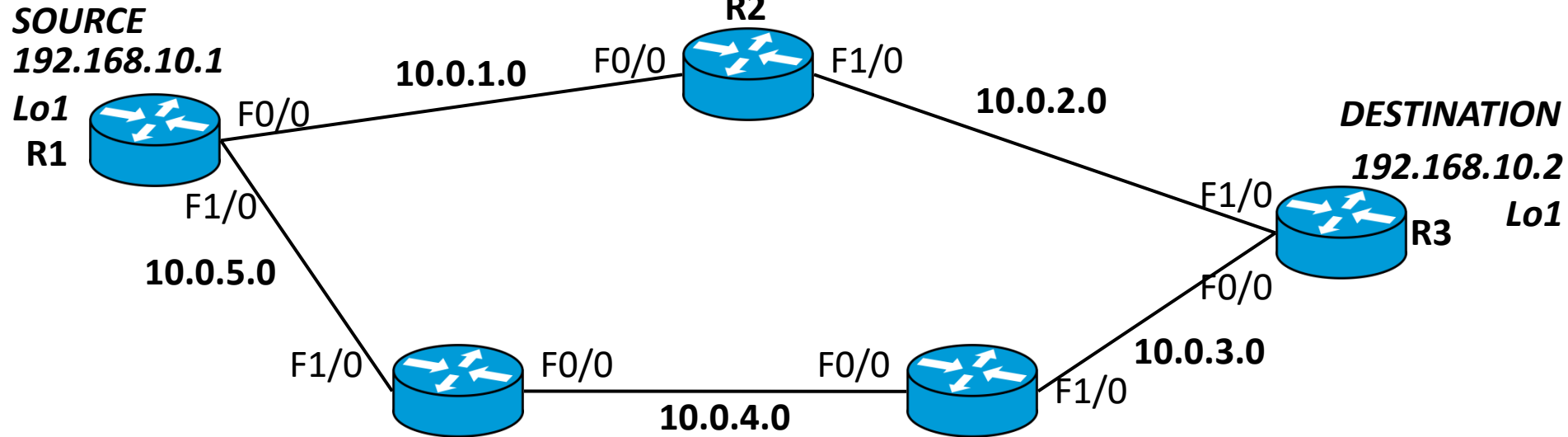
Loopback interface on R3 has a delay of 5000 us and 8000000 kb/s

for Path 1 is 5200 us and is 100000 kb/s

Cumulative Delay of Path 1 = 5200/10=520

Cumulative Bandwidth of Path 1 = $10^7 / 100000 = 100$

Route Metric of Path1 in EIGRP



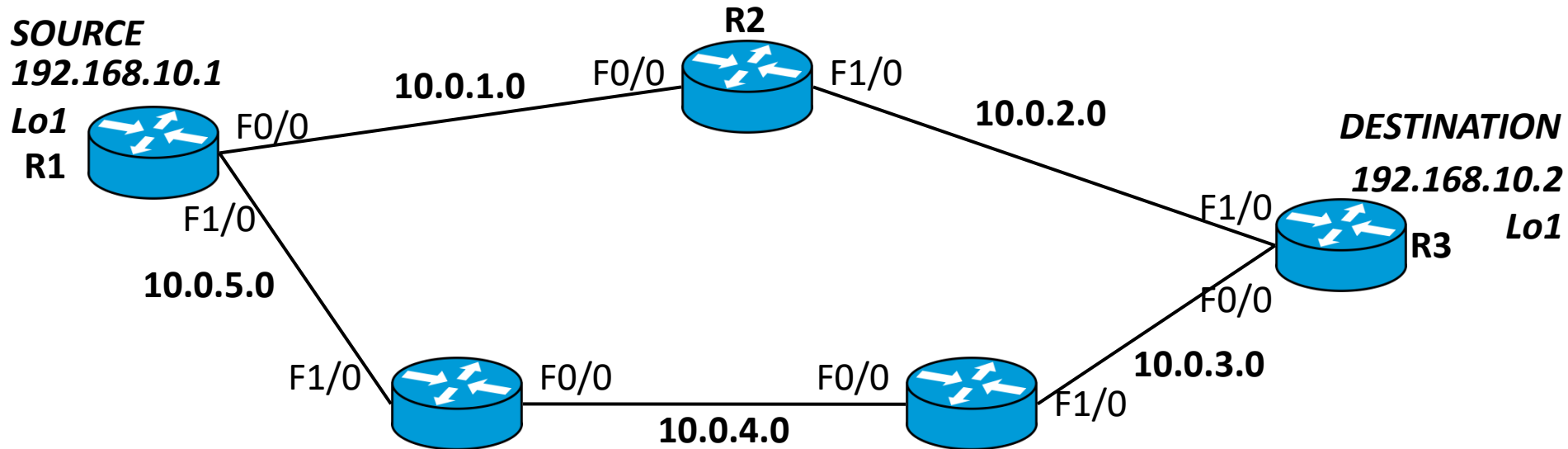
Route Metric (Path1) = $256 \times (\text{C. Delay} + \text{C. Bandwidth})$

Route Metric (Path1) = $256 \times (520 + 100)$

Route Metric (Path1) = 256×620

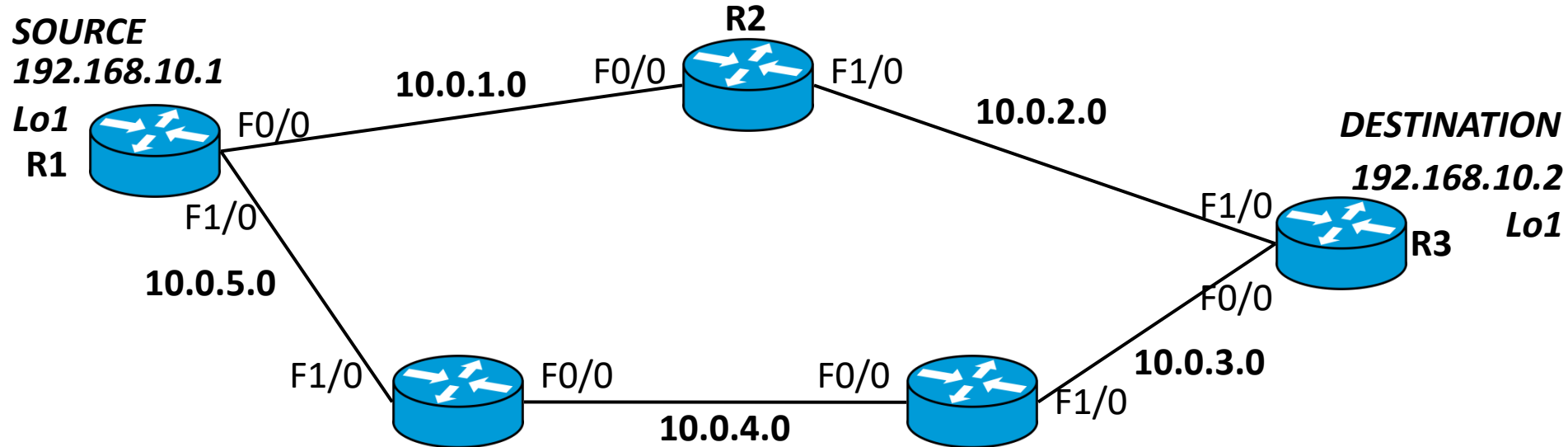
Route Metric (Path1) = 158720

Delay & B/W of Fastethernet and Loopback (Path2) in EIGRP



All fast ethernet ports has delay and B/w of 100 us and 100000 kb/s.
Loopback interface on R3 has a delay of 5000 us and 8000000 kb/s
Total Delay for Path 2 is 5300/10 us or 520 and Least Bandwidth is 100000 kb/s

Delay & B/W of Fastethernet and Loopback (Path2) in EIGRP



All fast Ethernet ports has delay and B/w of 100 us and 100000 kb/s.

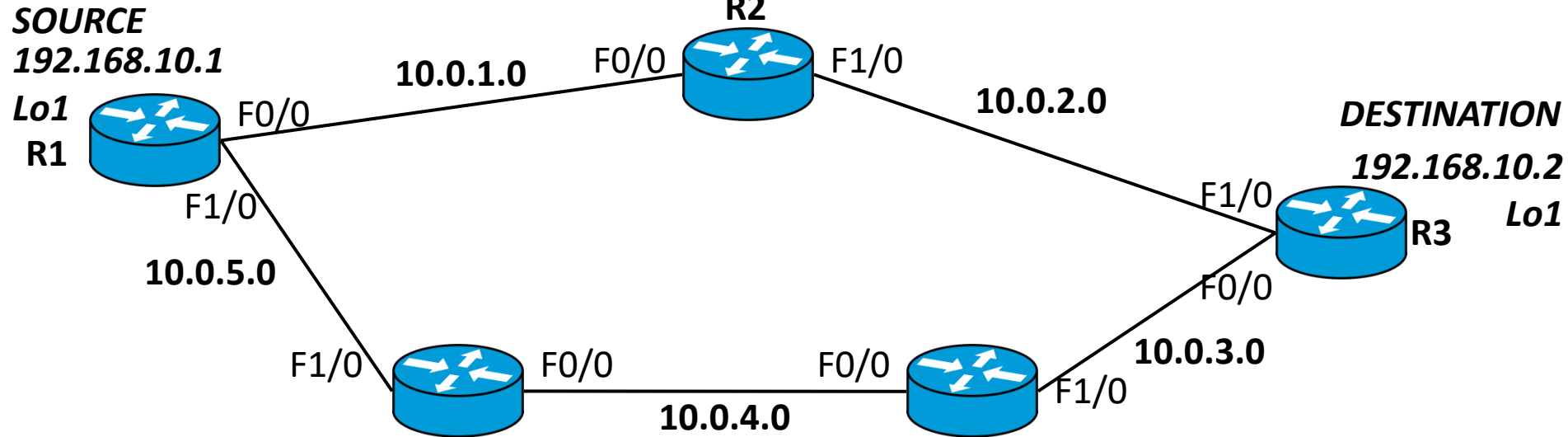
Loopback interface on R3 has a delay of 5000 us and 8000000 kb/s

for Path 2 is 5300 us and is 100000 kb/s

Cumulative Delay of Path 2 = 5300/10=530

Cumulative Bandwidth of Path 1 = $10^7 / 100000 = 100$

Route Metric of Path1 in EIGRP



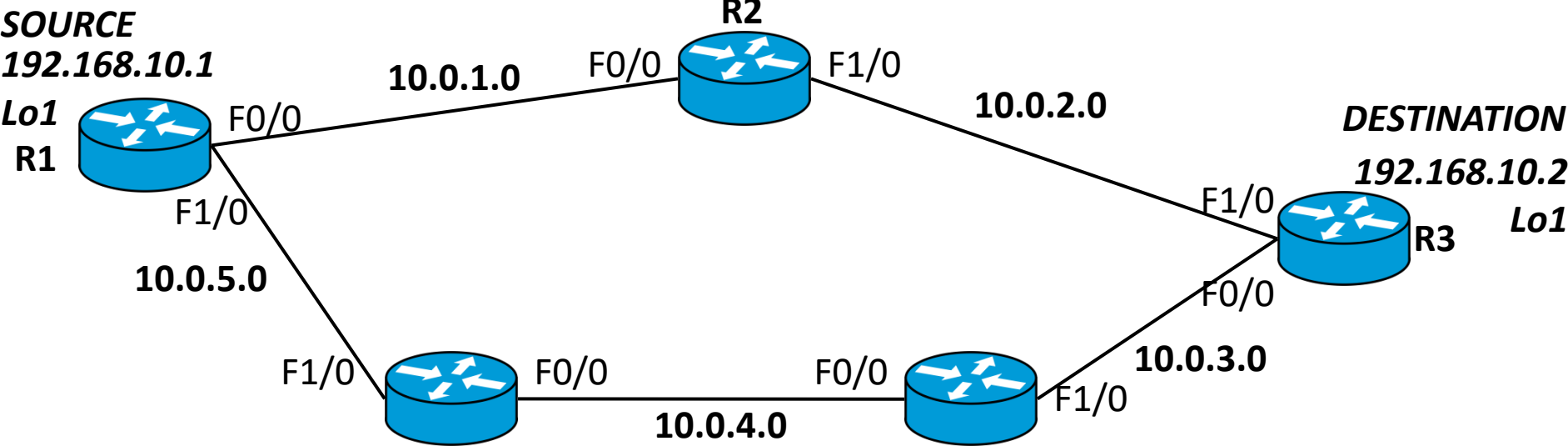
Route Metric (Path 2) = $256 \times (\text{C. Delay and C. Bandwidth})$

Route Metric (Path 2) = $256 \times (530 + 100)$

Route Metric (Path 2) = 256×630

Route Metric (Path 2) = 161,280

Route Metric of Path 1 vs Path 2



Route Metric of Path 1: 158720
Route Metric of Path 2: 161280

Final Statement: Path 1 having Low route metric is best path